



ISHFAQ AHMAD DAR

AI / ML Engineer

ABOUT ME

AI/ML Engineer with an **M.Sc. in Information Technology** and hands-on experience developing machine learning, computer vision, and Generative AI projects. Skilled in **Python, PyTorch, Vision Transformers (ViT), Explainable AI (Grad-CAM), FAISS, and Scikit-learn**, with practical experience building intelligent image retrieval systems, deep learning applications, and data-driven solutions. Passionate about solving real-world problems and seeking an **AI/ML Engineer** or **Computer Vision Engineer** role to apply and expand my technical expertise.

AI / ML PROJECTS

Content-Based Medical Image Retrieval (CBMIR) (Flagship Project)

GitHub: github.com/DarAshfaqDev/Content-Based-Medical-Image-Retrieval-CBMIR

- Developed an explainable Content-Based Medical Image Retrieval (CBMIR) system that retrieves visually similar medical images using deep feature embeddings extracted from a pre-trained Vision Transformer (ViT).
- Implemented a complete retrieval pipeline including image preprocessing, feature extraction, FAISS indexing, cosine similarity search, and ranked retrieval of similar medical images.
- Integrated Grad-CAM and Explainable AI (XAI) techniques to visualize image regions contributing to retrieval results, improving model interpretability.
- Built an interactive Gradio interface that allows users to upload medical images, retrieve similar cases, and visualize model explanations.

Text-to-SQL Agent using Gemini AI

GitHub: github.com/DarAshfaqDev/Text_to_SQL_Agent

- Developed a full-stack AI application that converts natural language questions into SQL queries using the Gemini API.
- Built a Flask REST API that validates AI-generated SQL queries before executing them on a SQLite database.
- Implemented schema-aware prompting to improve SQL generation accuracy and restricted execution to safe **SELECT** queries.
- Developed a React-based user interface for submitting natural language queries and displaying generated SQL statements with query results.

Weather Analytics & Forecasting Platform (in Progress)

GitHub: github.com/DarAshfaqDev/Weather-Analytics-Forecasting-Platform

- Developed an end-to-end weather analytics platform that collects real-time weather data, stores historical records, and performs data analysis using Python.
- Built machine learning models for short-term weather forecasting and implemented anomaly detection for identifying unusual weather patterns.
- Designed an interactive Streamlit dashboard to visualize weather trends, forecasts, and city-wise comparisons.
- Organized the application into modular components for data collection, preprocessing, model training, forecasting, and dashboard deployment.

Object Classification by Unsupervised Learning using VGG16 + KMeans

GitHub: github.com/Dar-Ishfaq-1/Object-Classification-by-Unsupervised-Learning-with-VGG16-KMeans

CERTIFICATIONS

Data Analytics Certification — TuteDude · 2024 | Power BI & SQL Certifications — Simplilearn · 2024 | AIML Engineering Certificate Course |

Data Analytics & Visualization — Accenture (Forage) · 2024 | Python Course — Mimo · 2023

CONTACT

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ai-and-ml-portfolio.vercel.app

EDUCATION

MSc Information Technology

University of Kashmir

2023 – 2025

BSc Non-Medical

Govt. Degree College, Kashmir

2019 – 2022

SKILLS

- Python, SQL, C++
- PyTorch, TensorFlow, Scikit-Learn
- Vision Transformers (ViT), CNNs
- Grad-CAM, FAISS Similarity Search
- LLM APIs (Gemini), Prompt Engineering
- Flask, React, Docker, Git
- Pandas, NumPy, EDA

LANGUAGES

English — Professional

Urdu — Native

Kashmiri — Native

RESEARCH INTERESTS

Computer Vision · Explainable AI

Retrieval-Augmented Systems

Generative AI · Medical AI